IMPERIAL COLLEGE LONDON

**EE4-51**
**EE9-FPN2-11**

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART IV: MEng and ACGI

# POWER SYSTEM ECONOMICS

Tuesday, 28 April 10:00 am

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**


Examiners responsible First Marker(s) : G. Strbac
Second Marker(s) : B.C. Pal



**© Imperial College London**

**Power System Economics – exam questions**

**Question 1**

**(a)** Describe the differences between electricity and other market commodities.

[2]

**(b)** Explain the notion of supply curve, demand curve, market clearing price and social welfare.

[2]

**(c)** The electricity demand $Q$ (kWh) in a given period is a function of price $\pi$ (£/kWh) and is given by the following expression: $\pi = -0.01Q + 9.2$ [£/kWh]. The marginal cost of production in the period under consideration is: $MC = 0.015Q + 1.5$ [£/kWh]

**(i)** Calculate the level of consumption, price, consumer surplus, producer surplus, and the total social welfare at the equilibrium.

[2]

**(ii)** At the price $\pi = 6.5$ £/kWh, calculate the consumption, consumer surplus, producer surplus, and the total social welfare at the equilibrium. How do these compare with the equilibrium calculated in (i)?

[3]

**(iii)** To what extent does this producer need to artificially increase the price, with respect to the equilibrium price calculated in (i), in order to maximize its profit?

[3]

**(d)** Two generators are available to supply a demand of $D = 600\ [MW]$. The cost of generating power $C_i(P_i)$ corresponding to each of the generators $i$ is:

$$C_1(P_1) = 300 + 8.05 * P_1 + 0.2 * P_1^2\ [\frac{£}{h}]$$

$$C_2(P_2) = 565 + 8.45 * P_2 + 0.3 * P_2^2\ [\frac{£}{h}]$$

Calculate the optimal production of each generator, electricity price and profits of each generator.

[4]

**(e)** Assume that the maximum output limit of each generator is:

$$P_1^{max} = 350\ [MW]$$

$$P_2^{max} = 300\ [MW]$$

Determine the optimal production of each generator, electricity price and the profits of each generator. Based on the results, briefly explain the impact of considering these maximum output limits on the total cost of operating the system and the profits of the two generators.

[4]

**Question 2**

**(a)** List and explain key objectives of unbundling and liberalisation of electricity markets. As the marginal pricing is used in pool markets, explain why producers taking part in a pool market do not have any incentives to bid at a price different to their marginal cost of production.

[4]

**(b)** Consider a market for electrical energy that is supplied by two generating companies with the two corresponding cost functions:

$$C_A(P_A) = 35 \cdot P_A + 0.37 \cdot P_A^2 \ [\frac{£}{h}]$$
$$C_B(P_B) = 39 \cdot P_B + 0.25 \cdot P_B^2 \ [\frac{£}{h}]$$

The inverse demand curve for this market is estimated to be:

$$\pi = 125 - 1.7 \cdot D \ [\frac{£}{MWh}]$$

where $\pi$ and $D$ are the electricity price and demand respectively. Assuming perfect competition, calculate the electricity price, level of demand, production levels and the profits made by the generating companies.

[4]

**(c)** Assuming a Cournot model of competition:
(i) Based on the generating companies cost functions and inverse demand curve presented in question (b), form a table to calculate the Nash equilibrium (NE) point of this market, i.e. market price, demand quantity and profit of each company for different levels of productions for each of the companies, as presented in Table 2.1

Table 2.1 Key parameters for assessing Nash Equilibrium

| | |
|---|---|
| $D$ (Demand) | $\Omega_A$ (profit A) |
| $\Omega_B$ (profit B) | $\pi$ (Price) |

Consider range of productions of company A at 14MW, 15MW and 16MW and of company B at 15MW, 16MW and 17MW.

[6]

(ii) Based on the results, briefly explain the impacts of imperfect competition on the market outcome.

[2]

(iii) State the optimality conditions and evaluate the exact NE of this market. At the NE, calculate the market price, demand quantity and profit of each company.

[4]

**Question 3**

**Part I**: Consider a centralised market that consists of four generating units, each unit produces/sells electricity by bidding at its marginal cost. Table 1 shows their relevant characteristics. Assume that 270MW reserve is required to maintain security for all loading conditions.

Table 3.1. Marginal cost, maximum output and reserve capability of the generating units

| Generation Units | Marginal cost (£/MWh) | Max Power output capacity Pmax (MW) | Max reserve power capacity Rmax (MW) |
|---|---|---|---|
| 1 | 24 | 265 | 210 |
| 2 | 19.5 | 255 | 180 |
| 3 | 36.5 | 270 | 0 |
| 4 | 4.5 | 270 | 0 |

**(a)** Draw the amount of reserve that generating units 1 and 2 can provide as a function of the amount of energy that they produce. [2]

**(b)** Assuming that the system demand is 420 MW, determine: i) the optimal power produced, and reserve provided by each of the four generating units, ii) the electricity and reserve prices, and iii) the combined profit that generation unit 2 derives from the energy and reserve markets. [4]

**(c)** Determine system demand range at which generating unit 1 becomes the marginal generator? For such demand range, repeat the calculation of **(b)**.

[4]

**Part II**: Syldavia Power is considering building a plant with the following characteristics:

Table 3.2. Assumed data for the generation plant

| | |
|---|---|
| Investment cost | 1010 £/kW |
| Capacity | 450MW |
| Expected plant life | 30 years |
| Heat rate at rated output | 9600 Btu/kWh |
| Expected fuel cost | 1.15£/MBtu |
| Expected utilization factor | 0.85 |
| Expected average selling price | 33 £/MWh |

In order to access the business case of this investment, the following questions need to be answered:

**(a)** Calculate the investment cost, estimated annual energy production, annual production cost, annual revenue and annual profit. Calculate the total cash flow over the expected plant life. [5]

**(b)** Give the formula of calculating the net present value (NPV) and describe how the calculation of the internal rate of return (IRR) is related to NPV. If the minimum acceptable rate (MARR) of Syldavia Power is set at 16.2%, and the IRR corresponding to the cash flow stream is 16%, can the investment in this plant be justified? [2]

**(c)** Assuming that the plant was built and after 15 years of operation there were some changes in the system. Because of increased demand, the price of the fuel burnt by the plant increased to 2.4 £/MBtu. Moreover, the government has imposed an environmental tax of 0.8 £/MWh on the output of fuel plants. Under these conditions, what would be the marginal cost of production of this plant? At the same time, competitors have put into service more efficient plants that have depressed the average price of electrical energy to 23.55 £/MWh. What would be annual loss of this plant? [3]

**Question 4**

**(a)** Describe the significance of contracts for differences and financial transmission rights in trading energy in a centralised electricity market.

[4]

**(b)** Consider two areas, Borduria and Syldavia, which are linked via a transmission line that can carry $F_{max} = 400\,[MW]$. There are two Generating companies in Borduria and two generating companies in Syldavia, with their respective cost functions and maximum output limits:

Borduria Power (BP): $C_{BP}(P_{BP}) = 10 * P_{BP} + 0.005 * P_{BP}^2\,[\frac{£}{h}]$ , $P_{BP}^{max} = 600\,[MW]$

Borduria Gen (BG): $C_{BG}(P_{BG}) = 16 * P_{BG} + 0.005 * P_{BG}^2\,[\frac{£}{h}]$ , $P_{BG}^{max} = 850\,[MW]$

Syldavia Energy (SE): $C_{SE}(P_{SE}) = 13 * P_{SE} + 0.01 * P_{SE}^2\,[\frac{£}{h}]$ , $P_{SE}^{max} = 500\,[MW]$

Syldavia Supply (SS): $C_{SS}(P_{SS}) = 23 * P_{SS} + 0.01 * P_{SS}^2\,[\frac{£}{h}]$ , $P_{SS}^{max} = 1200\,[MW]$

Both Borduria and Syldavia also have retail companies (Borduria Retail and Syldavia Retail) that sell energy to end consumers in the corresponding areas. Demand in Borduria is $D_B = 500MW$ while demand in Syldavia is $D_S = 1500MW$.

Given that the power flow on the line is from Borduria to Syldavia and the line is congested, calculate the optimal generation dispatch, the locational marginal prices in the two areas and the congestion surplus. Assuming that the system market participants do not enter into any form of contracts among themselves or the Independent System Operator (ISO) for network access, calculate revenues and profits from production for the four generating companies and charges that Borduria and Syldavia Retails will pay.

[8]

**(c)** Assume now that the following contracts for differences have been arranged:

| Seller | Buyer | Amount [MWh] | Strike Price [€/MWh] | Delivery Point |
|---|---|---|---|---|
| Borduria Power | Borduria Gen | 200 | 19 | Borduria |
| Borduria Power | Syldavia Retail | 100 | 19 | Borduria |
| Borduria Power | Borduria Retail | 300 | 19 | Borduria |
| Borduria Gen | Borduria Retail | 300 | 21 | Borduria |
| Borduria Gen | Syldavia Energy | 100 | 33 | Syldavia |
| Borduria Gen | Syldavia Retail | 100 | 34 | Syldavia |
| Borduria Retail | Syldavia Retail | 100 | 35 | Syldavia |
| Syldavia Energy | Syldavia Retail | 600 | 37 | Syldavia |
| Syldavia Supply | Syldavia Retail | 600 | 38 | Syldavia |

Calculate revenues and expenses for individual companies associated with these contracts, for the locational marginal prices determined in (b). Determine the net revenues, expenses and profits accounting for the respective quantities associated with the physical production and consumption. What is the total income that the ISO will receive?

[4]

**(d)** Repeat the evaluation under (c) assuming now that the following FTRs, for using the Transmission link in the direction from Borduria to Syldavia, have been purchased from the ISO.

| Participant | Amount of FTR purchased [MW] |
|---|---|
| Borduria Gen | 200 |
| Borduria Retail | 100 |
| Syldavia Retail | 100 |

What is the ISO's net position, after paying out the FTR holders from the congestion surplus?

[4]